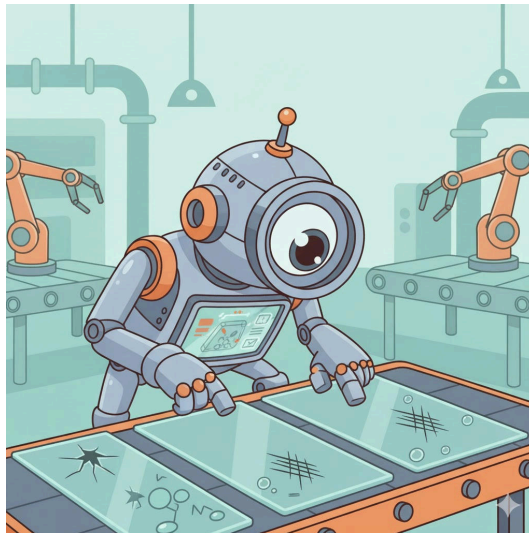


Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning



Bachelor Thesis

by
Florian Vollmer

Bachelors 2023

Submission date: 2026-07-02

Study Direction: Applied Digital Life Science

Supervisors:

Dr. Stefan Glüge

ZHAW Life Sciences und Facility Management, Wädenswil

Imprint

Recommended Citation:

Florian Vollmer (2026). *Automated Detection and Morphological Classification of Mast Cells in Bone Marrow Aspirates Using Deep Learning*

.

Keywords: Deep Learning, Object Detectors, Evaluation Metrics, Digital Lab and Production, Factory Production, Image Processing

Abstract

English Version

German Version

Acknowledgement

I would like to express my sincere gratitude to Dr. Stefan Glüge for setting up the project for this thesis and for his valuable support and guidance throughout the writing and compilation process of this thesis.

Table of Contents

List of Abbreviations	5
1. Introduction	6
1.1. Context and Motivation	6
1.2. Research Questions	6
1.3. Overall Concept	6
1.4. State of the Art	6
2. Methodology	7
2.1. Data Acquisition & Annotation	7
2.2. Data Preprocessing	7
2.3. Model Selection & Architecture	7
2.4. Model Training & Finetuning Strategy	7
2.5. Model Testing	7
2.6. Evaluation Metrics	7
2.7. UI-Creation	7
2.8. Implementation in USZ Infrastructure	7
3. Results	8
3.1. Model Results and Statistical Evaluation	8
3.2. Quantitative and Qualitative Evaluation	8
3.3. Evaluation Visualization	8
3.4. Real World Usage	8
4. Discussion	9
4.1. Interpretation of Results	9
4.2. Clinical Relevance	9
4.3. Error Analysis	9
5. Conclusion	10
5.1. Summary of Key Findings	10
5.2. Limitations and Challenges	10
5.3. Future Research Directions	10
6. Declaration on AI Assistance	11
7. References	12
8. Lists	13
List of Figures	13
List of Tables	13
9. Attachments	14

List of Abbreviations

- **AI:** Artificial Intelligence
- **AP:** Average Precision
- **CNN:** Convolutional Neural Network
- **COCO:** Common Objects in Context
- **CV:** Cross Validation
- **FN:** False Negative
- **FP:** False Positive
- **IoU:** Intersection over Union
- **mAP:** Mean Average Precision
- **RF-DETR:** Roboflow Detection Transformer
- **TN:** True Negative
- **TP:** True Positive
- **YOLO:** You Only Look Once

1. Introduction

1.1. Context and Motivation

1.2. Research Questions

1.3. Overall Concept

1.4. State of the Art

2. Methodology

2.1. Data Acquisition & Annotation

2.2. Data Preprocessing

2.3. Model Selection & Architecture

2.4. Model Training & Finetuning Strategy

2.5. Model Testing

2.6. Evaluation Metrics

2.7. UI-Creation

2.8. Implementation in USZ Infrastructure

3. Results

3.1. Model Results and Statistical Evaluation

3.2. Quantitative and Qualitative Evaluation

3.3. Evaluation Visualization

3.4. Real World Usage

Tests With Unseen Data

4. Discussion

4.1. Interpretation of Results

4.2. Clinical Relevance

4.3. Error Analysis

5. Conclusion

5.1. Summary of Key Findings

5.2. Limitations and Challenges

5.3. Future Research Directions

6. Declaration on AI Assistance

The author declares that the empirical research, data analysis, experimental design, and quantitative results presented in this thesis were generated independently and strictly remain the intellectual property of the author.

Artificial intelligence (AI) assistance was utilized solely for the purpose of language refinement, stylistic editing, and optimization of academic reading flow. Specifically, the AI model (Gemini 2.5 Pro) was employed to translate text, correct grammar, and restructure sentences into the required third-person past tense and formal academic style.

No core scientific content, model selection decisions, hyperparameter choices, or interpretation of results were delegated to or generated by the AI model.

7. References

[9] Gemini 2.5 Pro. (n.d.). Google Gemini. Retrieved October 1, 2025, from <https://gemini.google.com/app>

[10] DeepL Translate: The world's most accurate translator. (n.d.). <https://www.deepl.com/>

[11] F. Vollmer, „GitHub Repository: Glass Defect Detection-Evaluating-Object-Models“, GitHub, 1. December 2025. <https://github.zhaw.ch/vollmflo/Glass-Defect-Detection-Evaluating-Object-Detection-Models>

8. Lists

List of Figures

Cover Picture[9]

List of Tables

9. Attachments